Tableau Prep Data Cleaning Exercise Report

1. Dataset Import

The dataset tableau_prep_cleaning_exercise.csv was imported into Tableau Prep Builder for profiling, cleaning, transformation, and validation.

2. Dataset Profiling

The dataset was profiled before cleaning to identify data quality issues.

Profiling Activities Performed

- Checked missing values
- Verified data types
- Analyzed value distributions
- Detected duplicate rows
- Checked column cardinality
- Identified inconsistent text formats
- Validated emails and numeric values

Issues Found

Issue Type	Description
Missing Values	Missing values were found in Email, City, Order_Date, Ship_Date, Unit_Price, Age, and Satisfaction_Score fields
Data Type Issues	Unit_Price and date fields required datatype corrections
Duplicates	Duplicate rows were identified in the dataset
Inconsistent Text	Country, City, Product_Category, Customer_Name, and Payment_Method contained inconsistent capitalization and spelling
Invalid Patterns	Invalid email formats and inconsistent dates were detected

3. Column Analysis

Columns with High Cardinality

Column	Reason
Order_ID	Unique transaction identifier
Customer_ID	Unique customer identifier
Email	Mostly unique customer attribute
Product_ID	Product identifier

Potential Identifier Columns

- Order_ID
- Customer_ID
- Product_ID

Validation Checks

Validation Type	Result
Dates	Mixed date formats identified
Emails	Invalid email formats detected
Categories	Inconsistent capitalization corrected
Numeric Values	Invalid ranges filtered

4. Structure Analysis

Order ID Validation

Order_ID was checked to determine whether it uniquely identifies records. Duplicate rows and repeated orders were analyzed.

Customer ID Validation

Customer_ID consistency was checked against customer information.

Product Validation

Product_ID values were checked to ensure they matched consistent Product_Name values.

Calculated Field Validation

Sales calculations were validated using:

Expected_Sales = Quantity × Unit_Price × (1 − Discount)

5. Cleaning Rules Applied

Missing Values

Column Type	Action Taken
Categorical Fields	Replaced with “Unknown”
Notes	Replaced with “No Notes”
Invalid Rows	Excluded where necessary

A strict data-cleaning strategy was applied. Rows containing missing values in Email, City, Order_Date, Ship_Date, Unit_Price, Age, and Satisfaction_Score were excluded to retain only complete and validated records.

Duplicate Handling

- Exact duplicate rows were removed.

Invalid Values

Problem	Action
Invalid Emails	Removed
Invalid Numeric Values	Filtered
Inconsistent Text	Standardized

6. Cleaning and Transformation Steps

The following transformations were completed in Tableau Prep:

- Missing value handling
- Duplicate removal
- Data type correction
- Text standardization
- Email validation
- Numeric field validation
- Creation of calculated fields
- Category/bin creation

Calculated Fields Created

Field Name	Formula
Expected_Sales	Quantity × Unit_Price × (1 − Discount)
Age_Group	Categorized customer age groups
Sales_Category	Categorized sales amounts

7. Validation of Cleaned Dataset

Validation Checks

Validation	Status
Missing Values Checked	Completed
Duplicate Removal Verified	Completed
Data Types Verified	Completed
Numeric Ranges Validated	Completed
Calculated Fields Verified	Completed

Final Dataset Summary

Metric	Value
Original Rows	100
Final Rows	29
Total Rows Removed	71

Duplicate Rows Removed	5

8. Export Process

The cleaned dataset was exported from Tableau Prep as:

- tableau_prep_cleaned_dataset.csv

The Tableau Prep workflow was saved as:

- tableau_prep_cleaning_flow.tfl

9. ETL vs ELT Discussion

ETL

In ETL (Extract → Transform → Load), data profiling and cleaning occur before loading data into the warehouse.

ELT

In ELT (Extract → Load → Transform), raw data is loaded first and transformations are performed later inside the warehouse.

Recommended Approach

ETL is the better approach for this dataset because:

- The dataset is small and structured
- Data quality issues should be corrected before visualization
- Tableau analysis works better with pre-cleaned data
- Cleaning before loading reduces reporting errors

10. Conclusion

The dataset was successfully profiled, cleaned, validated, and transformed using Tableau Prep Builder. Data quality issues including missing values, duplicates, invalid formats, inconsistent text values, and invalid emails were corrected. After strict cleaning and filtering operations, the final cleaned dataset contained 29 validated records suitable for Tableau Desktop visualization and further analysis.